10 Ejercicios básicos con SSH

🔹 **Ejercicio 1 — Verificar si SSH está instalado**

Objetivo: comprobar cliente y servidor.

ssh -V

👉 Resultado esperado: versión de OpenSSH instalada.

```
└─$ ssh -V
OpenSSH_10.0p2 Debian-8, OpenSSL 3.5.2 5 Aug 2025
```

```
┌──(kali㉿kali)-[~]
└─$ ssh -V
OpenSSH_10.0p2 Debian-5, OpenSSL 3.5.0 8 Apr 2025
```

🔹 **Ejercicio 2 — Instalar y arrancar servidor SSH**

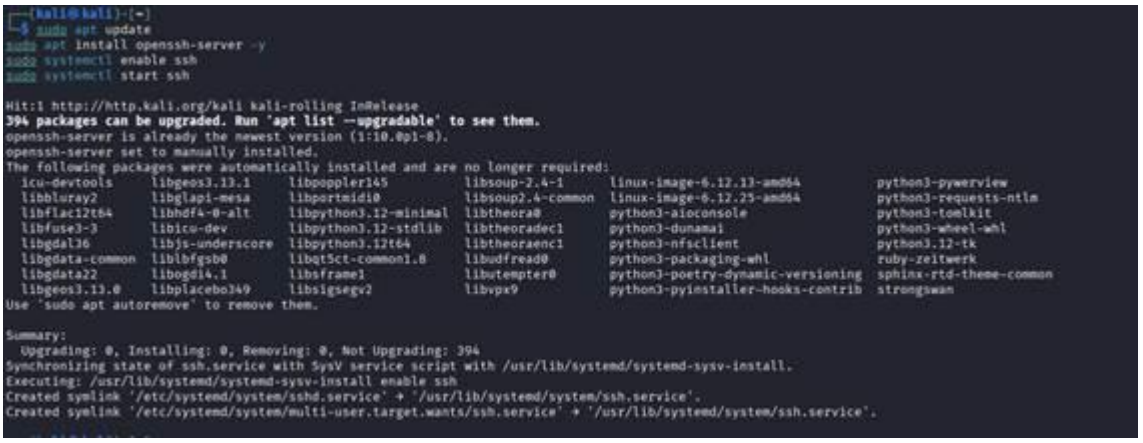
Objetivo: habilitar acceso remoto.

sudo apt update

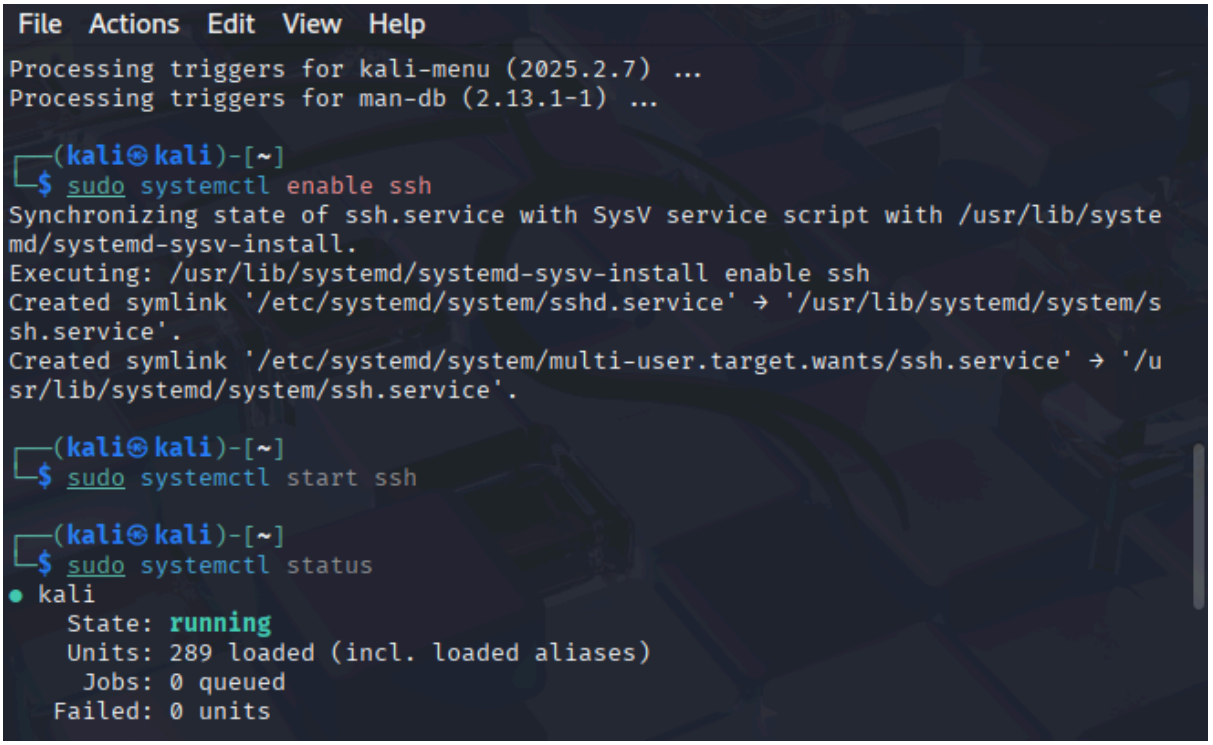
sudo apt install openssh-server -y

sudo systemctl enable ssh

sudo systemctl start ssh



👉 Resultado esperado: servicio SSH activo.

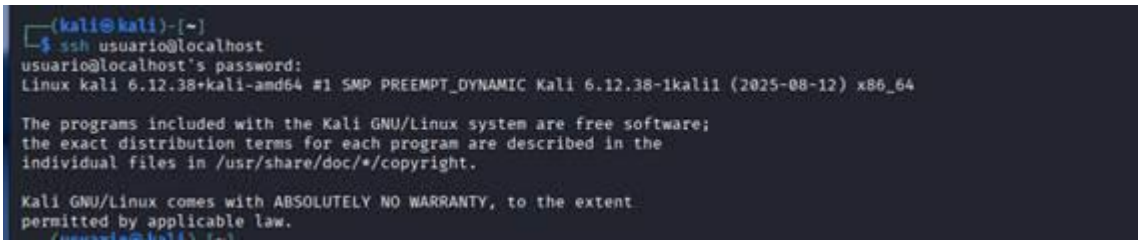


🔹 **Ejercicio 3 — Conectarse a localhost**

Objetivo: probar conexión básica.

ssh usuario@localhost

hay que crear previamente un usuario y crear una clave.



👉 Resultado esperado: pedir contraseña y entrar en sesión SSH.

```
┌──(usuario㉿kali)-[/home/kali]
└─$ ssh usuario@localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 key fingerprint is SHA256:DQ8St4vL3DkZ1LW/KMmOBgdxrk80DMr6qpRcwo8HohE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
usuario@localhost's password:
Last login: Mon Sep 29 07:54:06 EDT 2025 on pts/0
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-0
8-12) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
```

🔹 **Ejercicio 4 — Conectarse a otra máquina en la red**

Objetivo: practicar con dos VMs o un PC+móvil.

ssh usuario@192.168.1.50

👉 Resultado esperado: acceso remoto a otro host.

```
┌──(usuario㉿kali)-[~]
└─$ ssh usuario@192.168.1.50
ssh: connect to host 192.168.1.50 port 22: Connection refused
```

```
┌──(usuario㉿kali)-[~]
└─$ ssh usuario@192.168.0.24
The authenticity of host '192.168.0.24 (192.168.0.24)' can't be established.
ED25519 key fingerprint is SHA256:DQ8St4vL3DkZ1LW/KMmOBgdxrk80DMr6qpRcwo8HohE.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
    ~/.ssh/known_hosts:5: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.24' (ED25519) to the list of known hosts.
usuario@192.168.0.24's password:
Last login: Mon Sep 29 08:06:29 EDT 2025 from 192.168.0.24 on ssh
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-
```

ipftop mirar tu ip

```
┌──(usuario㉿kali)-[~]
└─$ ssh usuario@192.168.1.55
The authenticity of host '192.168.1.55 (192.168.1.55)' can't be established.
ED25519 key fingerprint is SHA256:zimehc2zuTjNtfNa5cc1FKbkgGsTIlUDZ+dJ23j5Bvc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.55' (ED25519) to the list of known hosts.
usuario@192.168.1.55's password:
Last login: Sat Sep 27 14:25:45 EDT 2025 from ::1 on ssh
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Sep 27 14:25:45 2025 from ::1
```

🔹 **Ejercicio 5 — Copiar un archivo con scp**

Objetivo: transferencia segura.

scp documento.txt usuario@192.168.1.50:/home/usuario/

hay que crear el documento.txt previamente

usuario

```
┌──(usuario㉿kali)-[~]
└─$ scp documento.txt usuario@192.168.1.55:/home/usuario/
usuario@192.168.1.55's password:
documento.txt
```

👉 Resultado esperado: archivo transferido al host remoto.

```
┌──(usuario㉿kali)-[~]
└─$ scp documento.txt usuario@192.168.0.24:/home/usuario/
usuario@192.168.0.24's password:
documento.txt                                 100%    0     0.0KB/s   00:00
```

🔹 **Ejercicio 6 — Copiar una carpeta con scp -r**

Objetivo: practicar recursividad.

scp -r carpeta usuario@192.168.1.50:/home/usuario/

👉 Resultado esperado: carpeta completa copiada.

```
┌──(usuario㉿kali)-[~]
└─$ scp -r carpeta usuario@192.168.1.55:/home/usuario/
usuario@192.168.1.55's password:
```

Hay que crear la carpeta previamente

```
┌──(usuario㉿kali)-[~]
└─$ mkdir carpeta

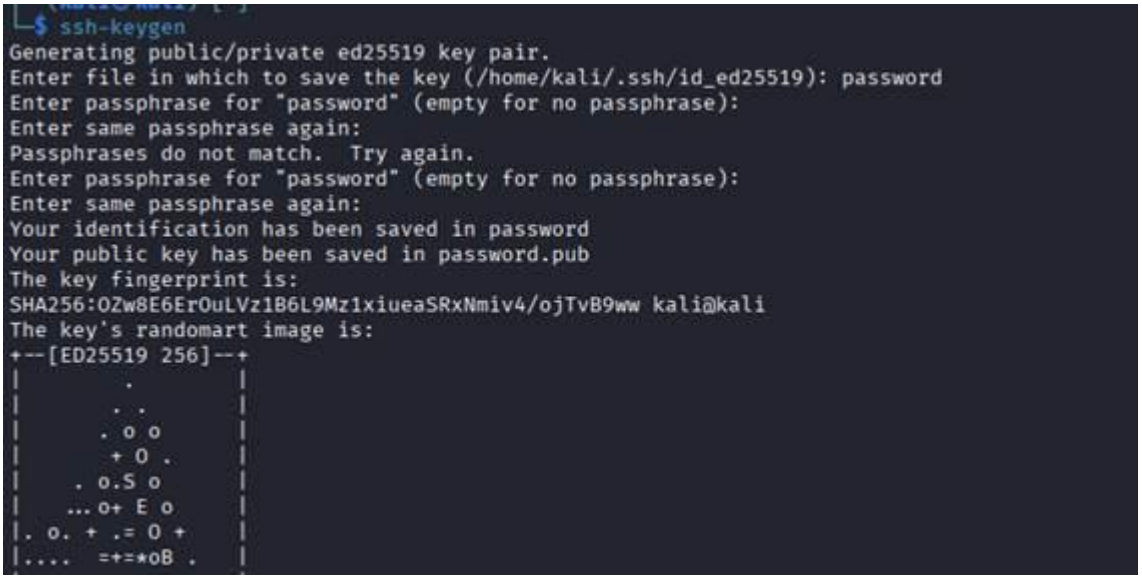
┌──(usuario㉿kali)-[~]
└─$ scp -r carpeta usuario@192.168.0.24:/home/usuario/
usuario@192.168.0.24's password:
```

🔹 **Ejercicio 7 — Usar autenticación con clave pública**

Objetivo: evitar contraseñas.

1. Generar clave:

   ssh-keygen



   Crear un fichero, yo lo he llamado password.pub

   Sino quieres crear un fichero le das al enter cuando salga el fichero por defecto id_ed25519

```
┌──(usuario㉿kali)-[~]
└─$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/usuario/.ssh/id_ed25519):
Enter passphrase for "/home/usuario/.ssh/id_ed25519" (empty for no passphrase):

Enter same passphrase again:
Your identification has been saved in /home/usuario/.ssh/id_ed25519
Your public key has been saved in /home/usuario/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:kaA2WrKUOOqrak/hEJ+syre9j5/BLLe6/L3Dk3WAGSQ usuario@kali
The key's randomart image is:
+--[ED25519 256]--+
|      .E..       |
| . . .  ..o      |
|o.+ =    o +     |
|.o+*..    + .    |
|..o=     S   .   |
|. + . o    . .   |
| o o . =. o .    |
|o.o.o + =        |
|Booo.BB* o+      |
+----[SHA256]-----+
```

2. Copiar clave al servidor:

   ssh-copy-id usuario@192.168.1.50

```
┌──(usuario㉿kali)-[~]
└─$ ssh-copy-id usuario@192.168.1.55
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/usuario/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
usuario@192.168.1.55's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'usuario@192.168.1.55'"
and check to make sure that only the key(s) you wanted were added.
```

```
┌──(kali㉿kali)-[~]
└─$ ssh-copy-id usuario@192.168.1.55
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed:
/usr/bin/ssh-copy-id: ERROR: No identities found
```

3. Conectarse sin contraseña.

- **Ejercicio 8 — Redirección de puerto con SSH (túnel simple)**

Objetivo: tunelizar una conexión.

ssh -L 8080:localhost:80 usuario@192.168.1.50

```
└─$ ssh -L 8080:localhost:80 usuario@192.168.1.55
Enter passphrase for key '/home/usuario/.ssh/id_ed25519':
Enter passphrase for key '/home/usuario/.ssh/id_ed25519':
Last login: Sat Sep 27 14:36:46 EDT 2025 from 192.168.1.55 on ssh
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Sep 27 14:36:47 2025 from 192.168.1.55
```

👉 Resultado esperado: acceder al puerto 80 del remoto desde http://localhost:8080.

```
┌──(usuario㉿kali)-[~]
└─$ ssh -L 8080:localhost:80 usuario@192.168.0.24
usuario@192.168.0.24's password:
Last login: Mon Sep 29 08:22:11 EDT 2025 from 192.168.0.24 on ssh
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-
08-12) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Sep 29 08:09:27 2025 from 192.168.0.24
┌──(usuario㉿kali)-[~]
└─$
```

🔹 **Ejercicio 9 — Ejecutar un comando remoto**

Objetivo: lanzar un comando sin abrir sesión completa.

ssh usuario@192.168.1.50 "uptime"

👉 Resultado esperado: salida del comando uptime del host remoto.

```
┌──(usuario㉿kali)-[~]
└─$ ssh usuario@192.168.0.24 "uptime"
usuario@192.168.0.24's password:
 08:26:47 up  1:57,  5 users,  load average: 0.16, 0.14, 0.16
```

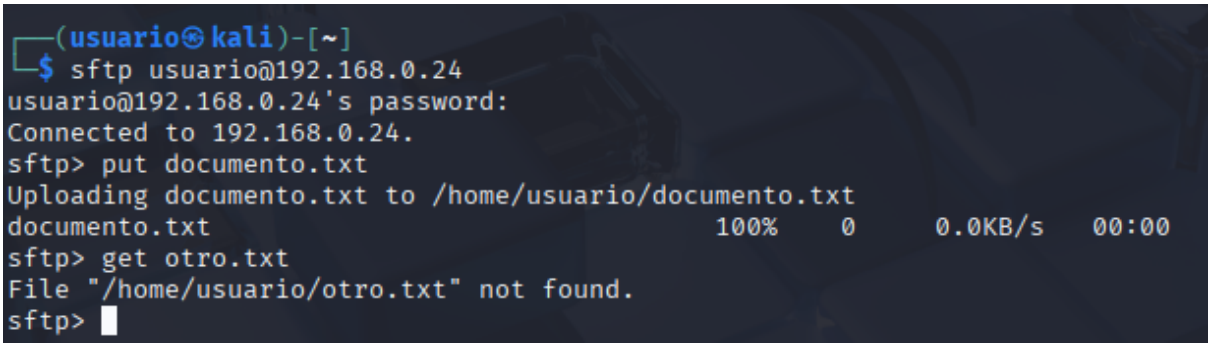
🔹 **Ejercicio 10 — Copia de archivos con sftp**

Objetivo: usar protocolo FTP seguro.

sftp usuario@192.168.1.50

sftp> put documento.txt

sftp> get otro.txt



Fallos al usar puerto 80

OPCIÓN 1: Iniciar sesión como root directamente

1. Sal del usuario actual:
2. exit
3. En tu terminal, inicia sesión como **root** si conoces la contraseña:
4. ssh root@192.168.1.55
5. Luego ejecuta:

ss -tuln | grep :80

instalar apache

apt update

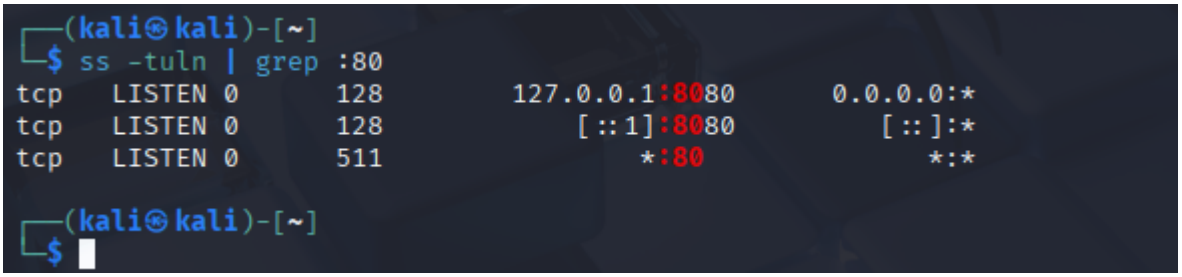
apt install apache2 -y

systemctl start apache2

ss -tuln | grep :80

Ahora deberías ver algo así:

LISTEN 0 128 0.0.0.0:80 0.0.0.0:*



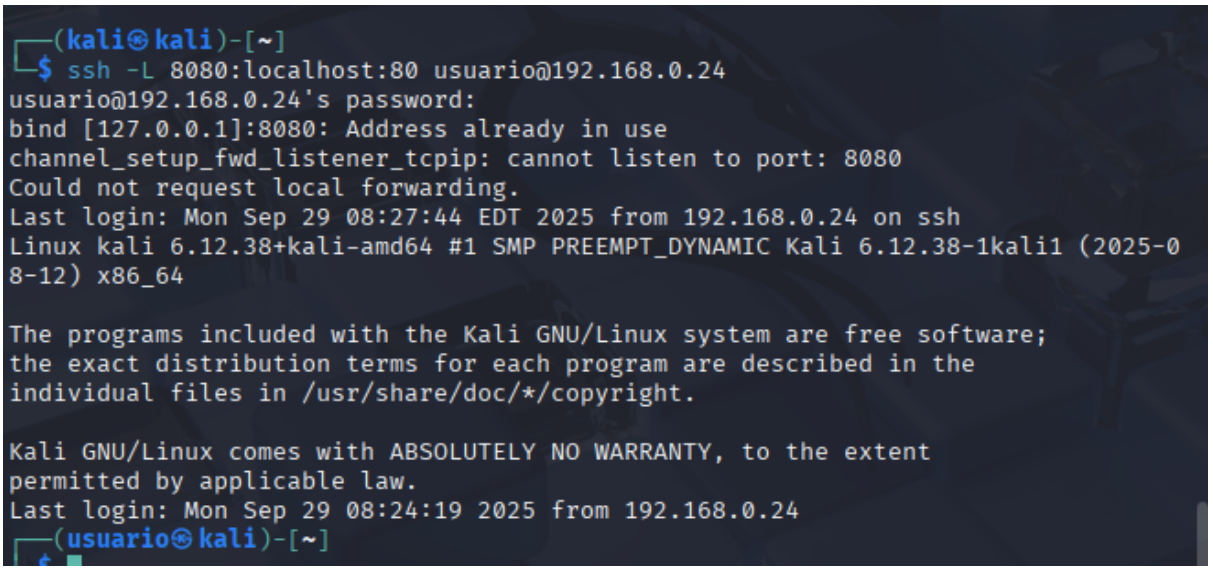
LISTEN 0 128 [::]:80 [::]:*

👉 3. En tu máquina cliente (desde la que haces el túnel SSH):

Ejecuta:

ssh -L 8080:localhost:80 usuario@192.168.1.55

(Sustituye usuario por el que tenga acceso, o usa kali si has iniciado sesión con ese.)



👉 4. Desde tu navegador o terminal en la máquina cliente:

En el navegador:

http://localhost:8080

O en terminal:

curl http://localhost:8080

```
┌──(usuario㉿kali)-[~]
└─$ curl http://localhost:8080

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.or
g/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Debian Default Page: It works</title>
    <style type="text/css" media="screen">
  * {
    margin: 0px 0px 0px 0px;
    padding: 0px 0px 0px 0px;
  }

  body, html {
    padding: 3px 3px 3px 3px;

    background-color: #D8DBE2;
```

```
/etc/apache2/
|-- apache2.conf
|       `--  ports.conf
|-- mods-enabled
|       |-- *.load
|       `-- *.conf
|-- conf-enabled
|       `-- *.conf
|-- sites-enabled
|       `-- *.conf
          </pre>
          <ul>
                      <li>
                         <tt>apache2.conf</tt> is the main configuration
                         file. It puts the pieces together by including all re
maining configuration
                         files when starting up the web server.
                      </li>
```

aparecerá la página de apache.